Histology of Vessels and Lymphatics

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Session Objectives

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- Students will be able to compare and contrast large, medium, and small arteries and veins.
- Students will be able to describe different capillary types and how their morphology varies according to their function.
- Students will be able to discuss the histology and function of lymphatic vessels.
- Students will be able to discuss the unique functions of endothelial cells.
- Students will be able to discuss microvascular components that regulate blood flow.

Source: Chapter 13, pp 400-408, 432-433. Cardiovascular System, Histology: A text and Atlas, Ross and Pawlina,

• GENERAL CONCEPTS: VESSELS

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Arteries and veins have three layers (tunics)

1. **Tunica intima**: Innermost layer of endothelium, basal lamina of the endothelial cells, and subendothelial layer (loose connective tissue). Arteries: subendothelial layer of intima contains a sheet-like layer (lamella) of fenestrated elastic material (the internal elastic membrane) which lets substances diffuse through and reach cells deep in the vessel wall.
2. **Tunica media**: Middle layer, mainly circumferentially arranged layers of smooth muscle cells. Arteries: this layer is pretty thick and extends from the internal to the external elastic membrane. The latter is a layer of elastin that separates the tunica media from the tunica adventitia.
3. **Tunica adventitia** – outermost connective tissue layer; contains (among other things) vasa vasorum which supply blood to vessel walls

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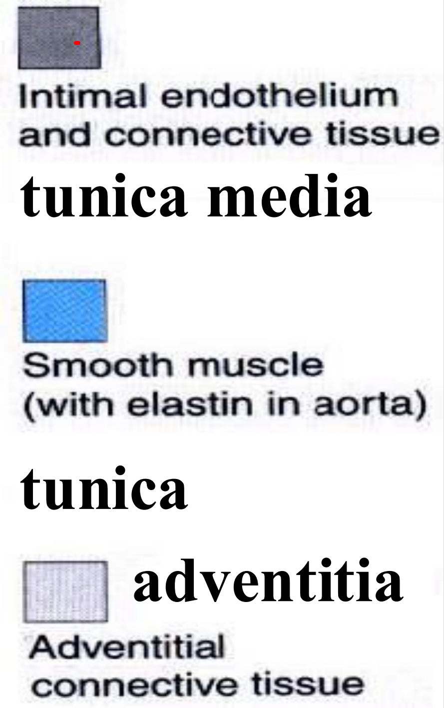
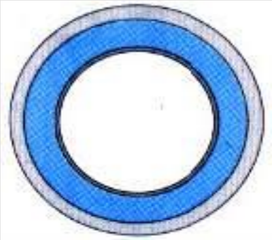
tunica intima



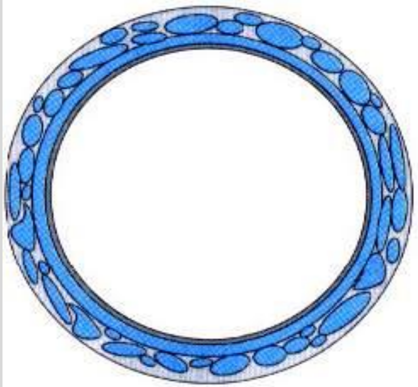
tunica media

tunica adventitia


Aorta


Muscular artery


Arteriole


Vena cava


Vein


Venule

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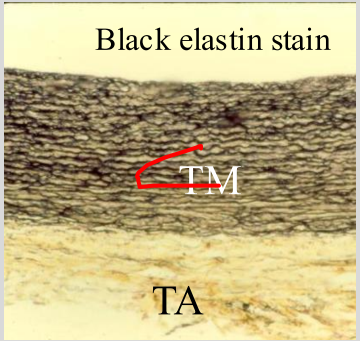
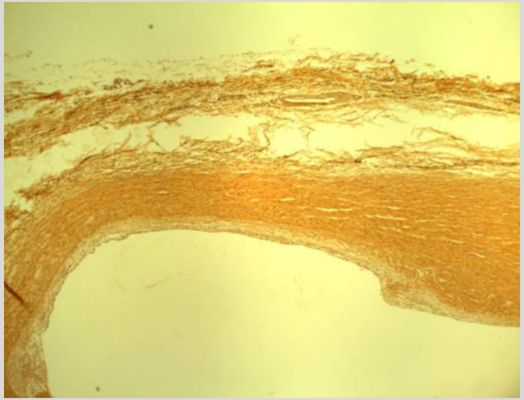
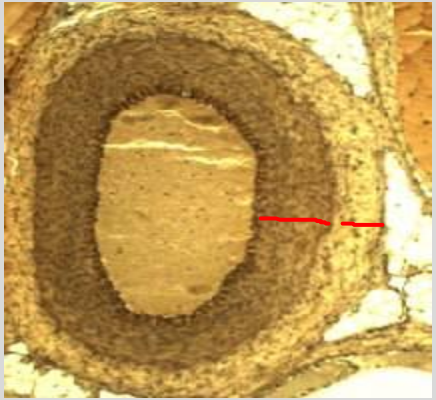
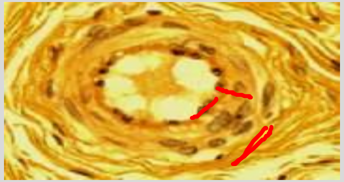




Aorta

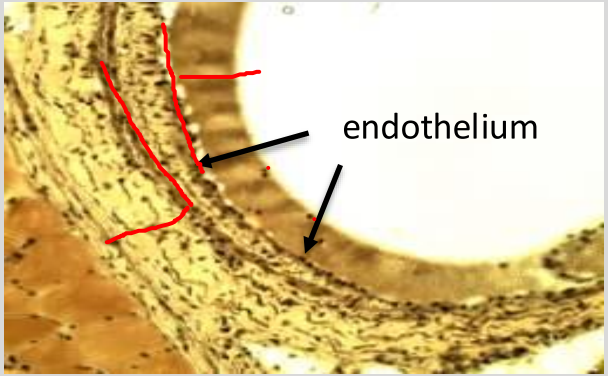


Muscular artery



Arteriole

Vena cava



Vein

Venule

Source: Gerdes

Basic differences between arteries and veins

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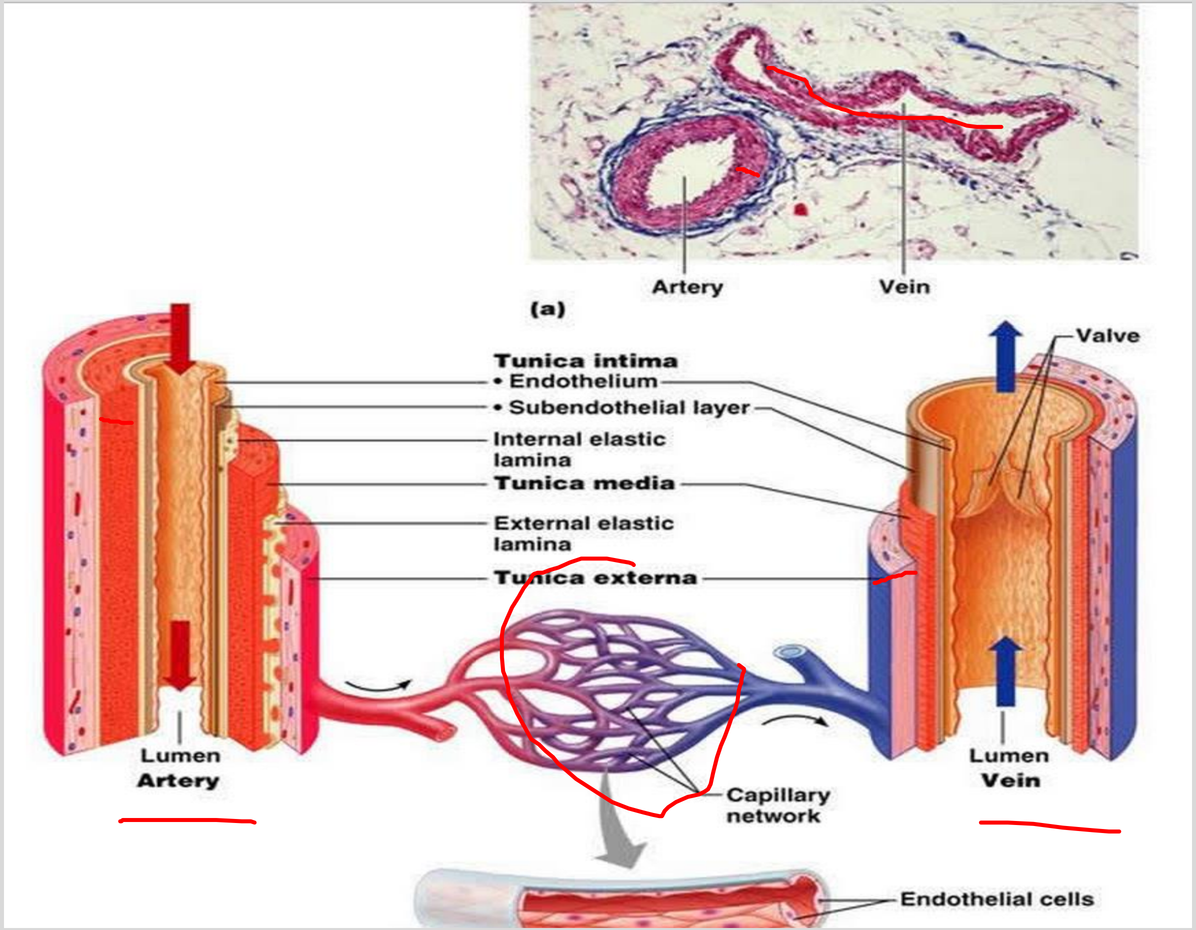
When comparing paired vessels:

– 1. Arteries have thicker walls and relatively thicker TM

– 2. Veins have thinner walls, larger and more collapsed lumen, and TA is relatively more prominent



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Source: http://classes.midlandstech.edu/carterp/Courses/bio211/chap19/chap19.html

ARTERIES

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- Large artery (elastic artery): > 1 cm dia
- Medium artery (muscular artery): 2-10 mm
- Small artery: 0.1 – 2 mm
- Arteriole: 5-30 µm, smooth muscle layer 1 -2 cells
- Capillary: 4-10 µm, only endothelium

Source: Ross and Pawlina

VEINS

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- Postcapillary venule: 10-50 µm, only endothelium and pericytes
- Muscular venule: 50-100 µm, smooth muscle layer 1 - 2 cells
- Small vein: 0.1 – 1 mm
- Medium vein - 1 – 10 mm
- Large vein > 1cm

Ross and Pawlina

Large Elastic Arteries

Tunica Intima

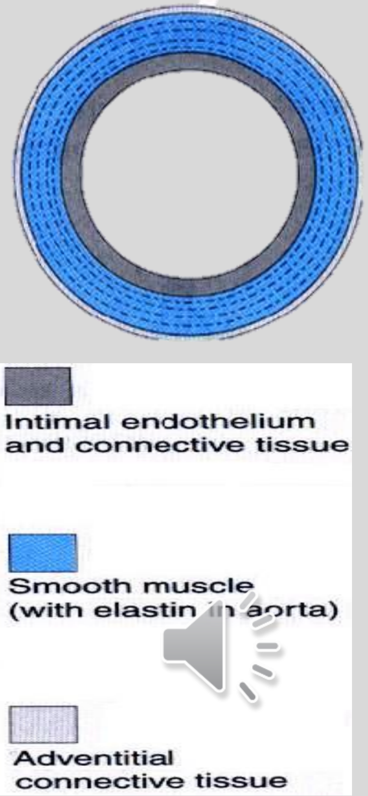
- **-endothelium:** simple squamous
- **-subendothelium:** CT & elastin
- **-internal elastic lamina:** elastin: fenestrated sheets

Tunica Media

- -smooth muscle – spindle shaped cells
- -collagen fibers and ground substance made from smooth muscle cells, not fibroblasts
- **-external elastic lamina**

Tunica Adventitia

- -collagen fibers, elastic fibers (not sheets)
- -fibroblasts and macrophages (main cells)
- -vasa vasorum and nervi vascularis



LARGE (ELASTIC) ARTERY

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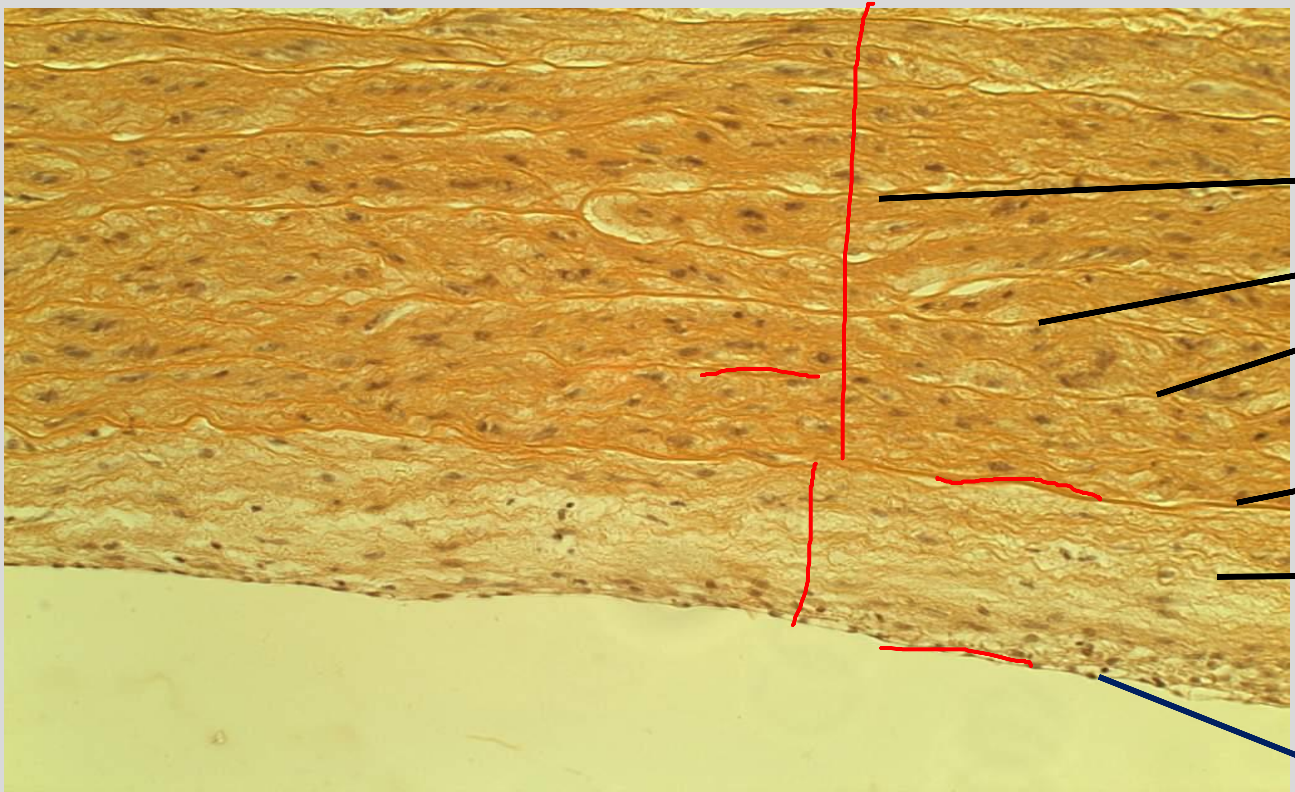

t. media
elastin sheets
50-70 in aorta

t. intima
internal elastic lamina

t. intima
subendothelium
connective tissue, elastin

Endothelium:
simple squamous

Source: Gerdes

Large (elastic) arteries

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- Examples include aorta and pulmonary arteries
- Conduct blood from heart; pressure generated by ventricular contraction (systole) moves blood through the elastic arteries and causes the wall of the large elastic arteries to distend.
- <u>During diastole</u>, the <u>recoil of the distended elastic arteries</u> serves to maintain the arterial blood pressure and flow of blood within the vessels

Tunica intima of elastic artery

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Endothelial lining with basal lamina: flat elongated cells with long axes oriented parallel to direction of blood flow in artery.

- Cells are joined by tight junctions (zonulae occludentes) and gap junctions.

Subendothelial layer of connective tissue

Internal elastic membrane (lamina)

<u>Muscular Arteries: most 'named' arteries of body</u>

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tunica intima

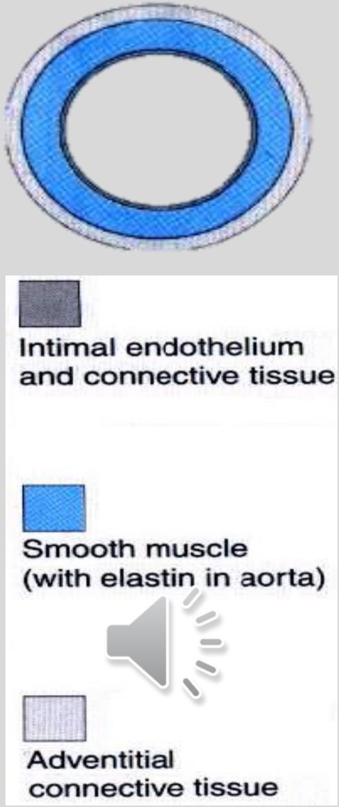
- **-endothelium**
- **-subendothelium:** sparse
- **-internal elastic lamina:** PROMINENT

tunica media

- **-thickest layer**
- **-smooth muscle: PREDOMINANT**
- **-collagen with little elastin**
- **-external elastic lamina:** in largest ones

tunica adventitia

- **-collagen fibers, elastic fibers**
- **-fibroblasts and adipose**
- **-vasa vasorum and nervi vascularis**



Muscular Artery

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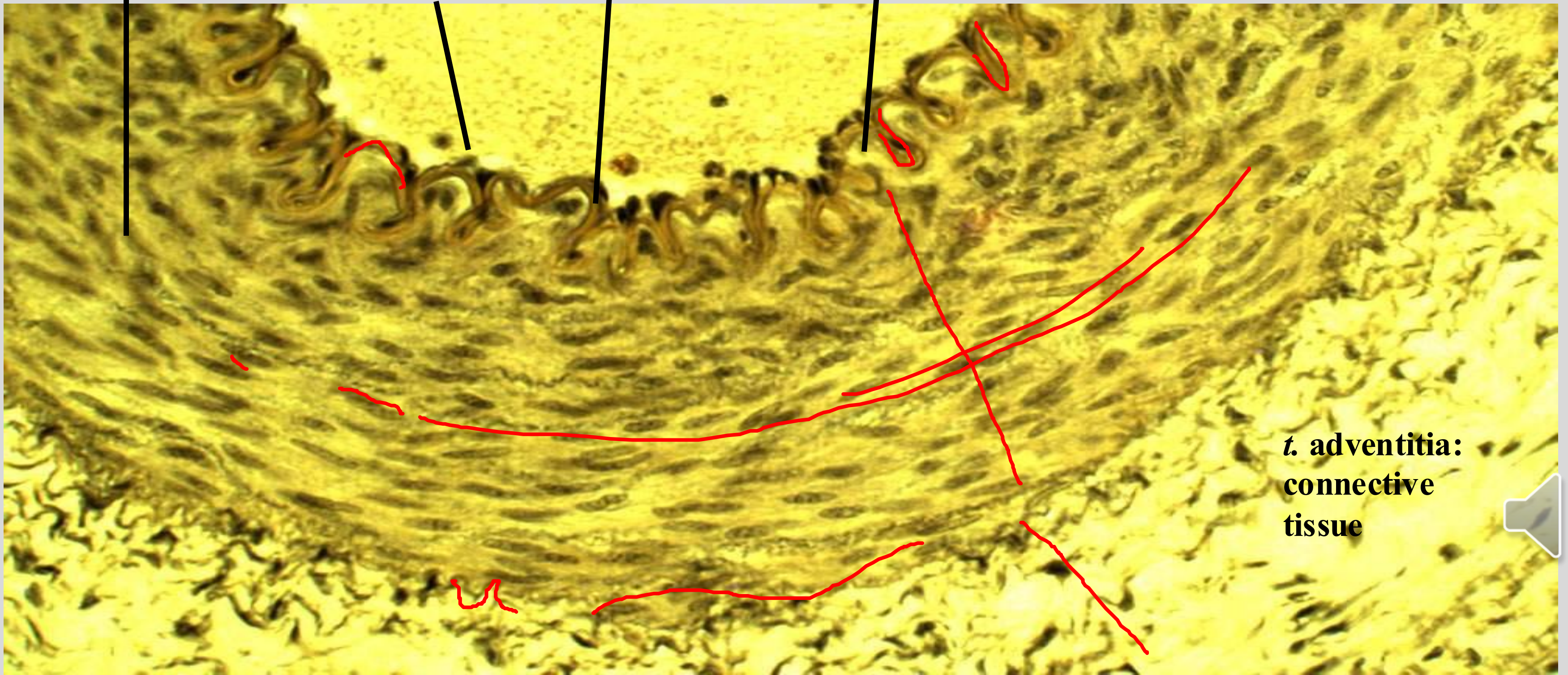
t. media: **smooth muscle**

t. intima: **simple squamous endothelium**

t. intima: **internal elastic lamina**

t. intima: **subendothelium** (sparse)



Small arteries to arterioles

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tunica intima:

Intimal endothelium and connective tissue

- **-endothelium**
- **-internal elastic lamina:** in larger ones

tunica media:

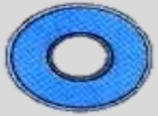
Smooth muscle (with elastin in aorta)

- **-smooth muscle:**
  - ~<u>8 layers in small artery</u>
  - <u>1-2 smooth muscle layers in arteriole</u>
- **-some collagen:** in largest small arteries

tunica adventitia:

Adventitial connective tissue

- **-connective tissue**



<u>arterioles regulate the flow of blood to capillaries</u>

<u>specialized arterioles can act as an A-V shunt</u>

Arteriolar Regulation

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- Precapillary sphincter: slight thickening of smooth muscle at origin of a capillary bed from an arteriole
- Arterioles can dilate up to 100% from resting diameter and can maintain up to 40% constriction for a long time
- Large decrease or increase in vascular resistance has direct effect on blood flow and systemic arterial pressure and this regulation can direct blood flow to where it is needed (e.g., during strenuous exercise, blood flow to skeletal muscle is increased by dilation of arterioles while blood flow to the intestine is reduced by arteriolar constriction)

CAPILLARIES

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- Allow fluids containing gases, metabolites, and waste products to diffuse through their thin walls
- Single layer of endothelial cells and their basal lamina
- Three types: continuous, fenestrated, and discontinuous

Capillaries

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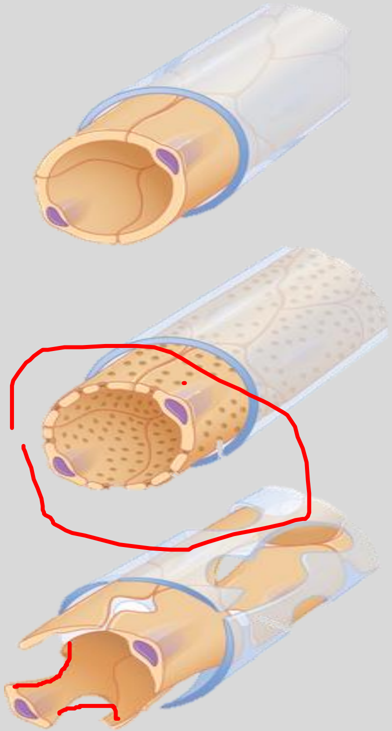
Tunica intima: **-one to three endothelial cells**
-basal lamina / basement membrane

3 types--

continuous pass metabolites and oxygen with tissues for waste and carbon dioxide (**muscle, lungs, CNS**); may have pericytes – see later discussion

fenestrated found in organs involved in absorption (intestine), filtration (kidney), and secretion (endocrine glands); have pinocytotic vesicles; pores/diaphragms

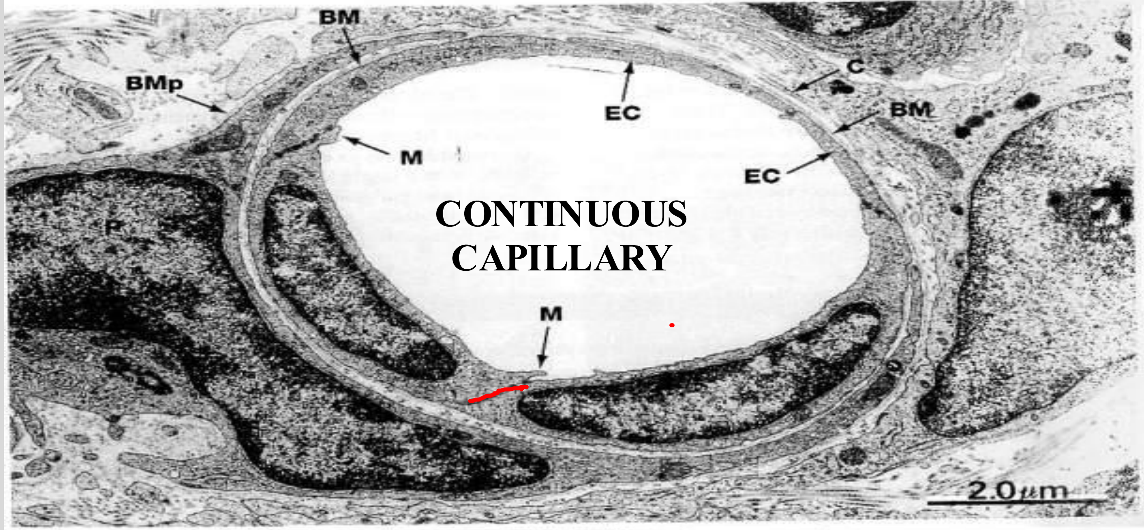
discontinuous a.k.a. **sinusoids** pass blood cells (liver, spleen, and bone marrow)

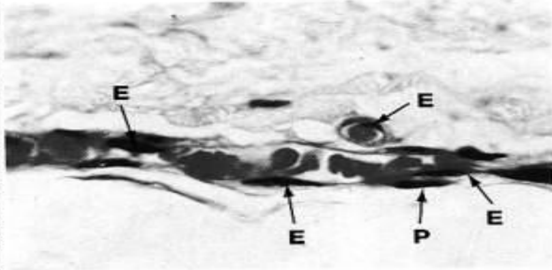


Source: Opentextbc.ca

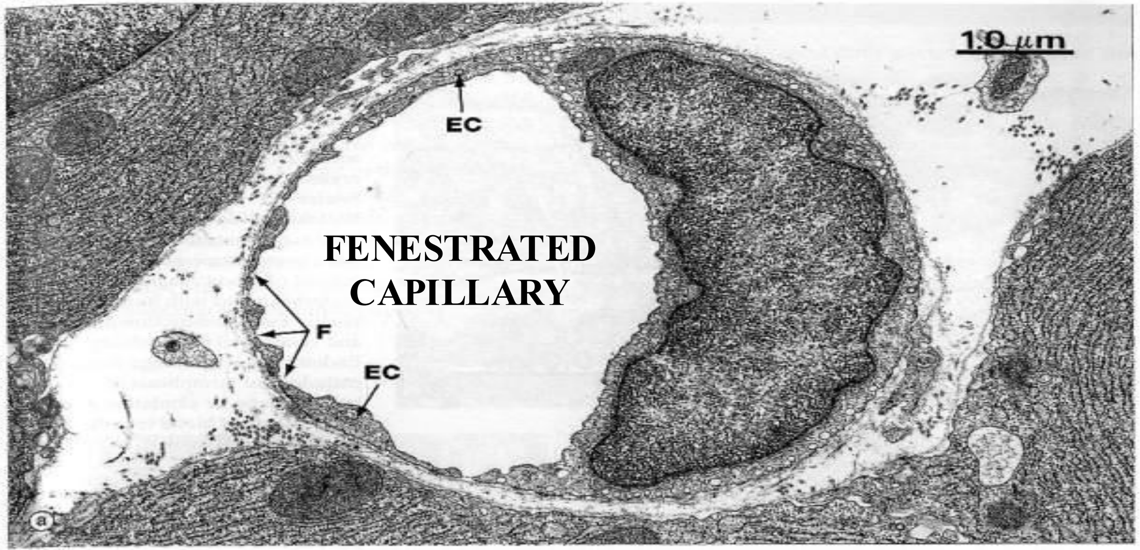
simple squamous endothelium [EC]
basement membrane [BM]

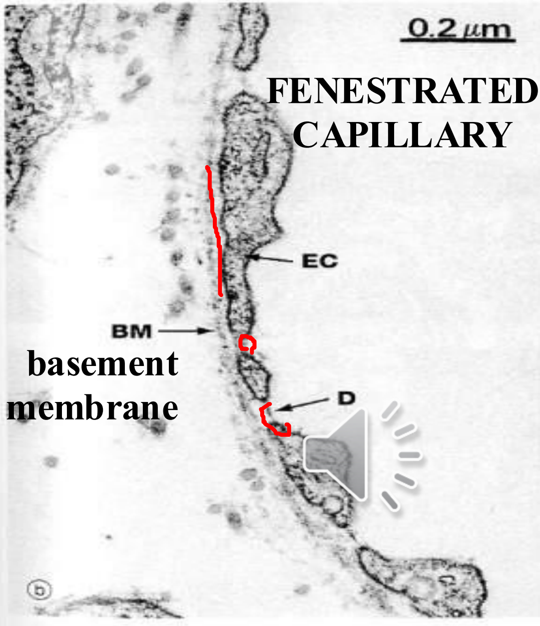
cellular junctions (marginal folds) [M]





Fenestrations [F] or pores with diaphragms [D] of radially oriented fibrils





Endothelial cells

- Maintain selective permeability barrier (simple diffusion, active transport, pinocytosis, receptor-mediated endocytosis)
- Maintain non-thrombogenic barrier
  - secrete anticoagulants, anti-thrombogenic agents and prothrombogenic agents – thrombomodulin, prostacyclin, TPA, antithrombin III, heparin, tissue thromboplastin, von Willebrand factor, etc.
- Modulate blood flow and vascular resistance via vasoconstrictors (e.g. Endothelin, ACE) and vasodilators (e.g. NO)
- Regulate cell growth (e.g. secrete VEGF, vascular endothelial growth factor)
- Regulate immune response, maintenance of extracellular matrix, and lipoprotein metabolism
- NOT JUST A STATIC TUBE LINER!

Ross Histology

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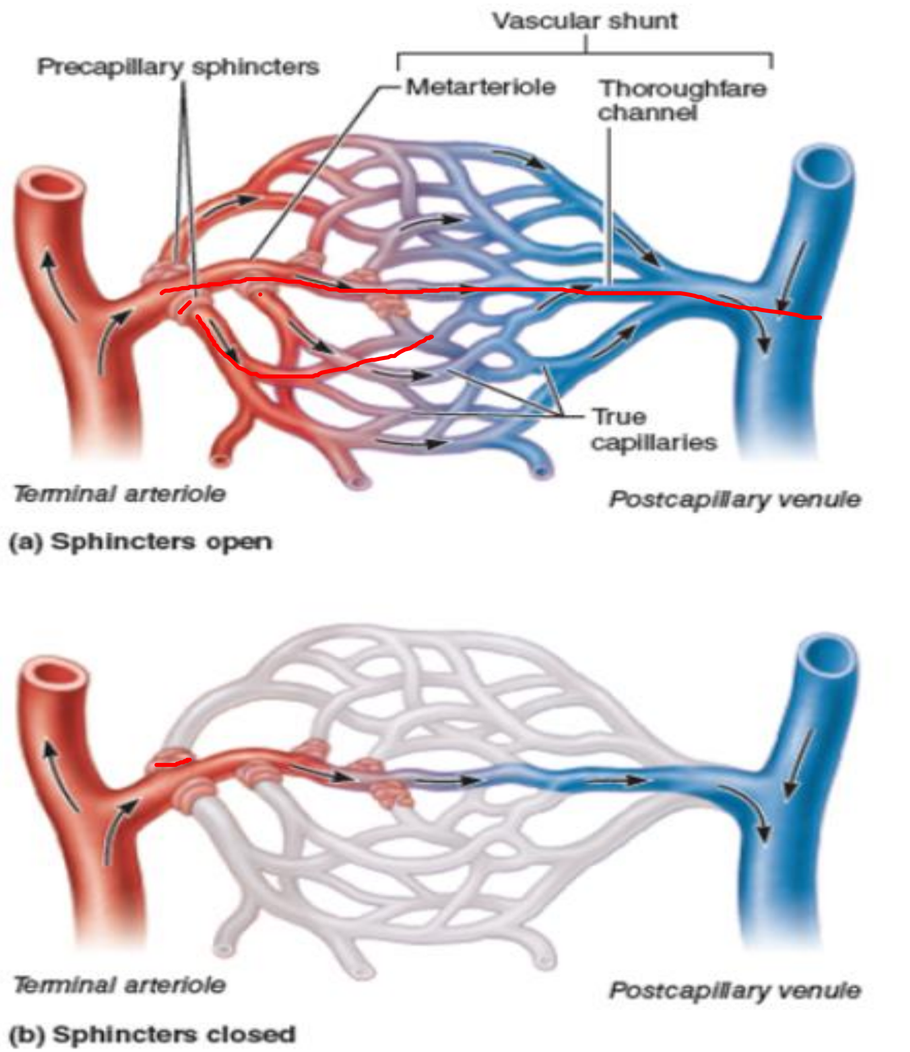
ARTERIOVENOUS (AV) SHUNTS

- Allow blood to bypass capillaries by providing direct routes between arteries and veins.
- AV Fistula is abnormal AV connection that can lead to volume overloading of the heart. A penetrating wound may pierce artery and vein and create connection.
- **Normal AV shunts**
- Preferential thoroughfares, proximal segment is known as a metarteriole, allow some blood to pass more directly from artery to vein (lack of precapillary sphincters at venous end).
- Commonly found in the skin of the fingertips, nose, lips, erectile tissue of penis and clitoris.
- Serve in thermoregulation at body surface (e.g. closing an AV shunt in skin causes blood to flow through the capillary bed, enhancing heat loss).

Normal AV shunting mechanism



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Source: chegg.com

<u>Post-capillary venules and small veins</u>

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- **Tunica intima – endothelium**
  - Postcapillary venules in lymph nodes are called high endothelial venules (HEVs) because of prominent cuboidal endothelial cells
- **Tunica media**
  - **1-2 layers of smooth muscle** in small veins;
  - <u>No tunica media present in postcapillary venules</u>
  - <u>Pericytes are present in post capillary venules</u>
- **Tunica adventitia - connective tissue**
  - <u>thin layer in small veins (not post capillary venules</u>)
  - Respond to **histamine** (released by mast cells, ↑ capillary permeability) and **serotonin** (mostly vascular contraction)

Pericytes

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- In postcapillary venules and some continuous capillaries, **pericytes** may be associated with the endothelium. They surround the venule or capillary, with branching cytoplasmic processes, and are enclosed by basal lamina which is continuous with that of the endothelium
- Derived from the same precursor cells that forms endothelial cells in new vessels,
- Pericytes can give rise to endothelial and smooth muscle cells during vessel growth (e.g, wound healing)
- Contractile, can respond to NO

Pericytes

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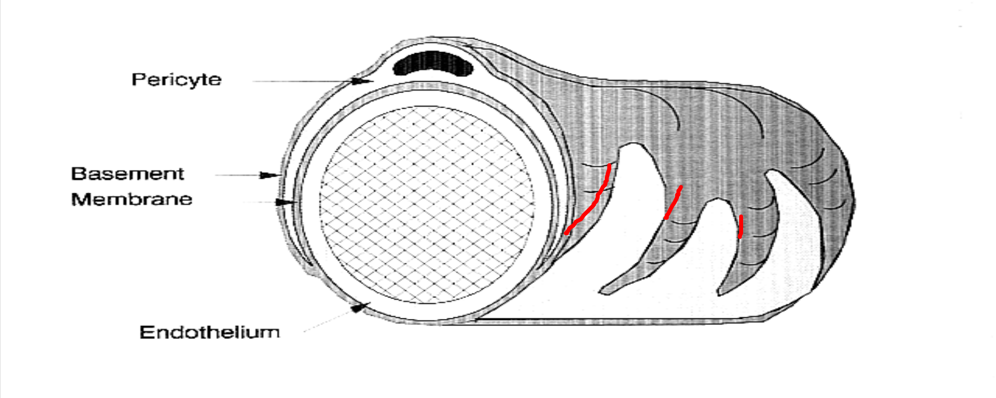

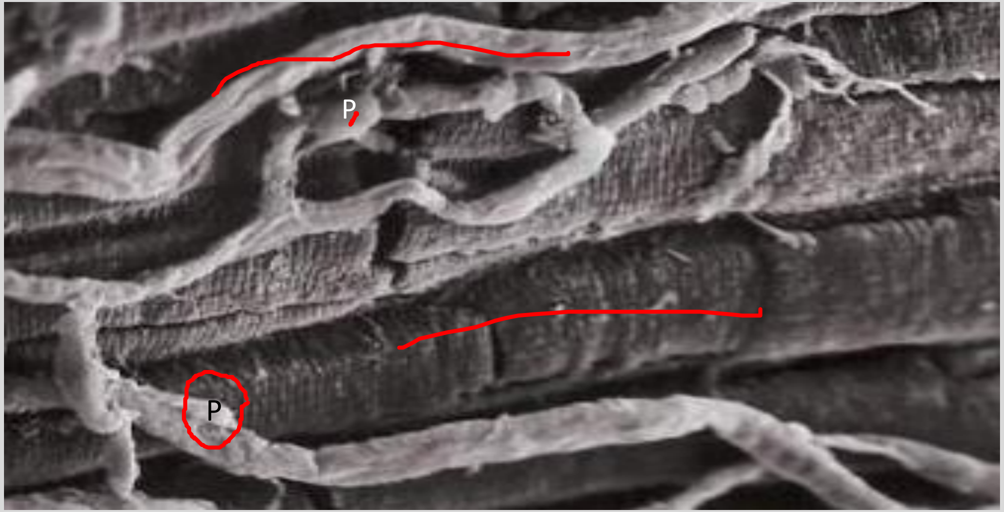

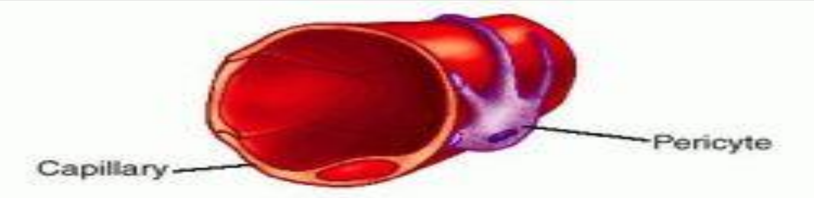

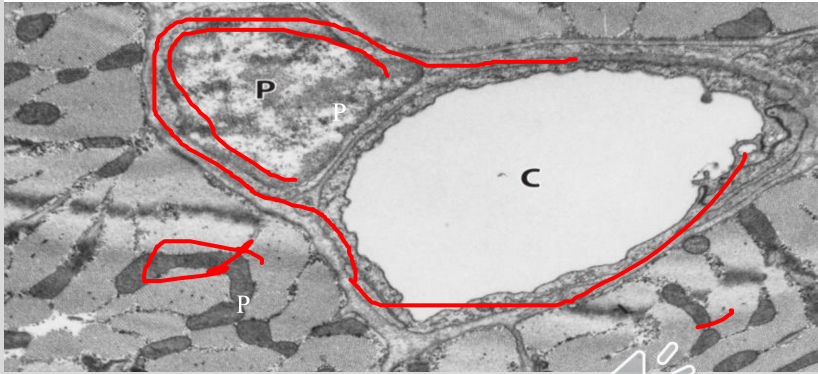

Above: TEM showing pericyte adjacent to capillary in dog heart.
Left: Scanning EM of pericytes (P) running along capillaries in rat heart.

Source: AM Gerdes

Medium veins

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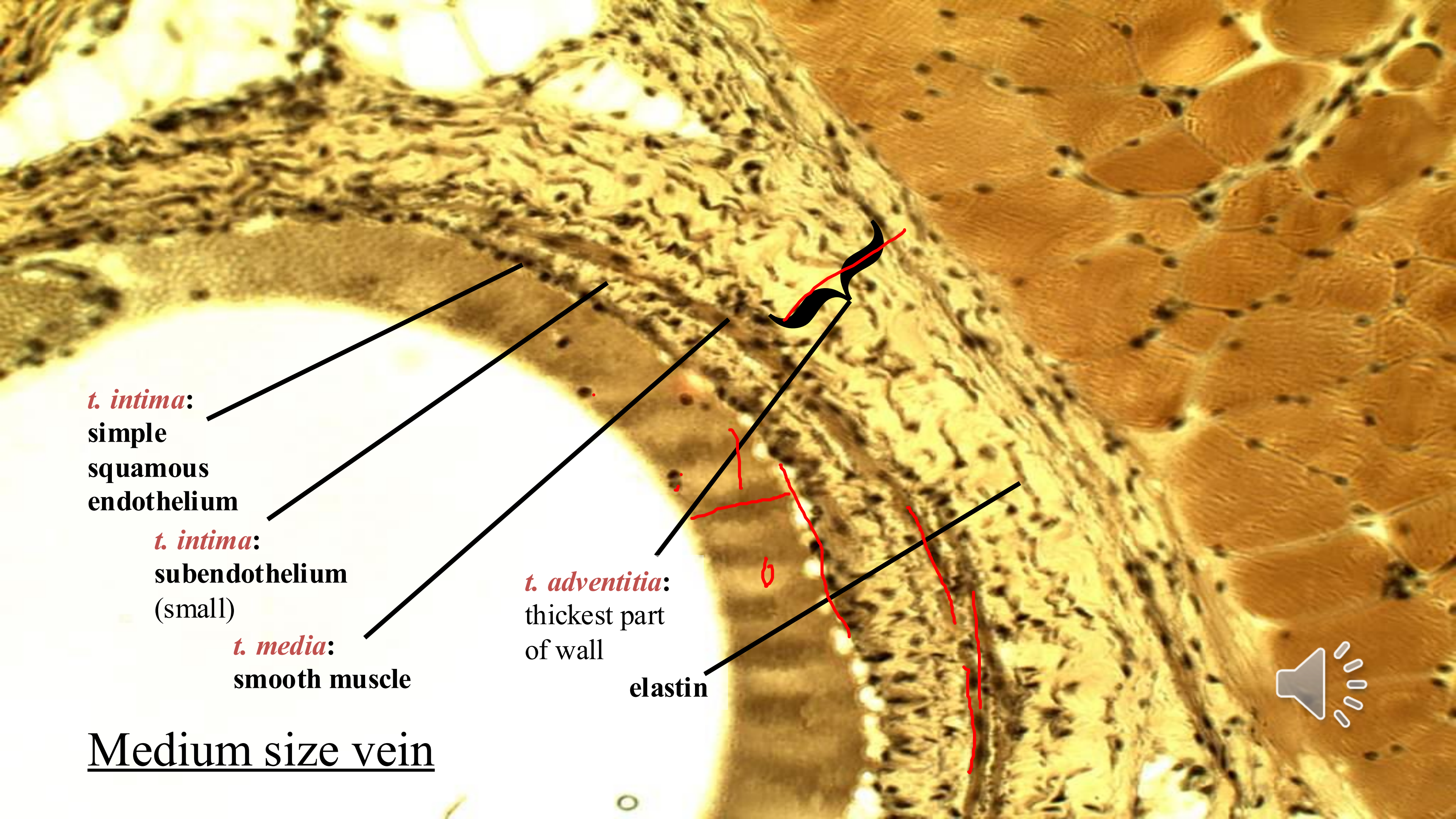
Most ‘named’ veins in body; valves characteristic, especially in legs

- **Tunica intima – endothelium**
  - **subendothelium** – thin, with occasional smooth muscle cells
- **Tunica media - smooth muscle (thinner than medium sized arteries)**
  - **collagen and fibroblasts**
- **Tunica adventitia - thickest layer in medium veins**
  - **elastic fibers longitudinal collagen & smooth muscle**


t. intima: simple squamous endothelium
t. intima: subendothelium (small)
t. media: smooth muscle
t. adventitia: thickest part of wall
elastin
Medium size vein

Large veins

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- **Tunica intima - endothelium**
  - thin subendothelium; boundary between tunica intima and media not often clear
- **Tunica media**
  - relatively thin with mostly circular smooth muscle, fibroblasts, and collagen
- **Tunica adventitia -** thickest layer in large veins (e.g subclavian veins and venae cavae)
  - longitudinal smooth muscle, fibroblasts, collagen, and elastic fibers

Note: Gravity can have effect on vein structure- Larger veins in head have thinner walls. Saphenous vein in leg has thicker muscle layer and is preferred for coronary bypass surgery because it is already more “muscularized”.

Large Vein (cross section)

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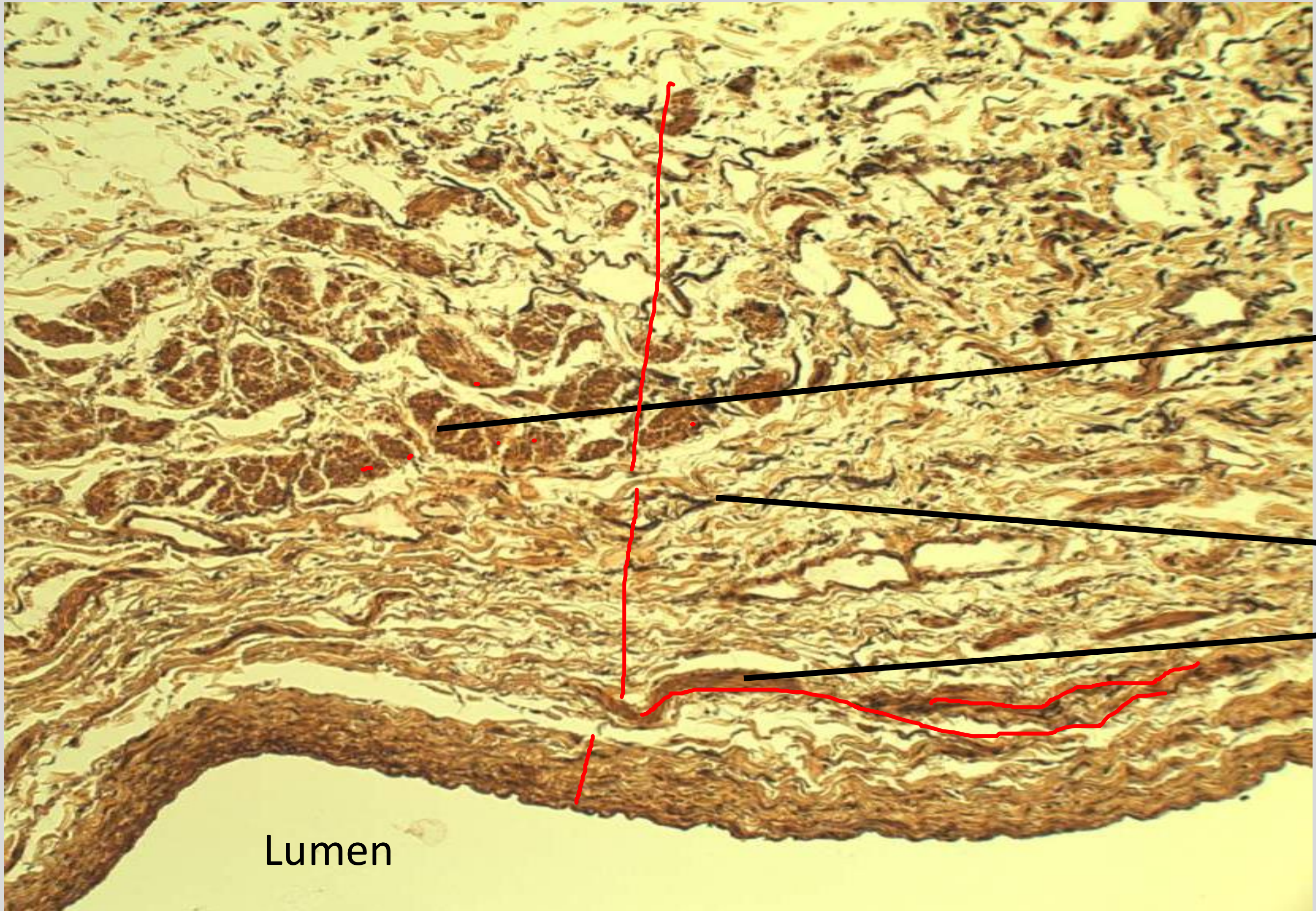


t. adventitia: **longitudinal smooth muscle** (cut in cross-section)

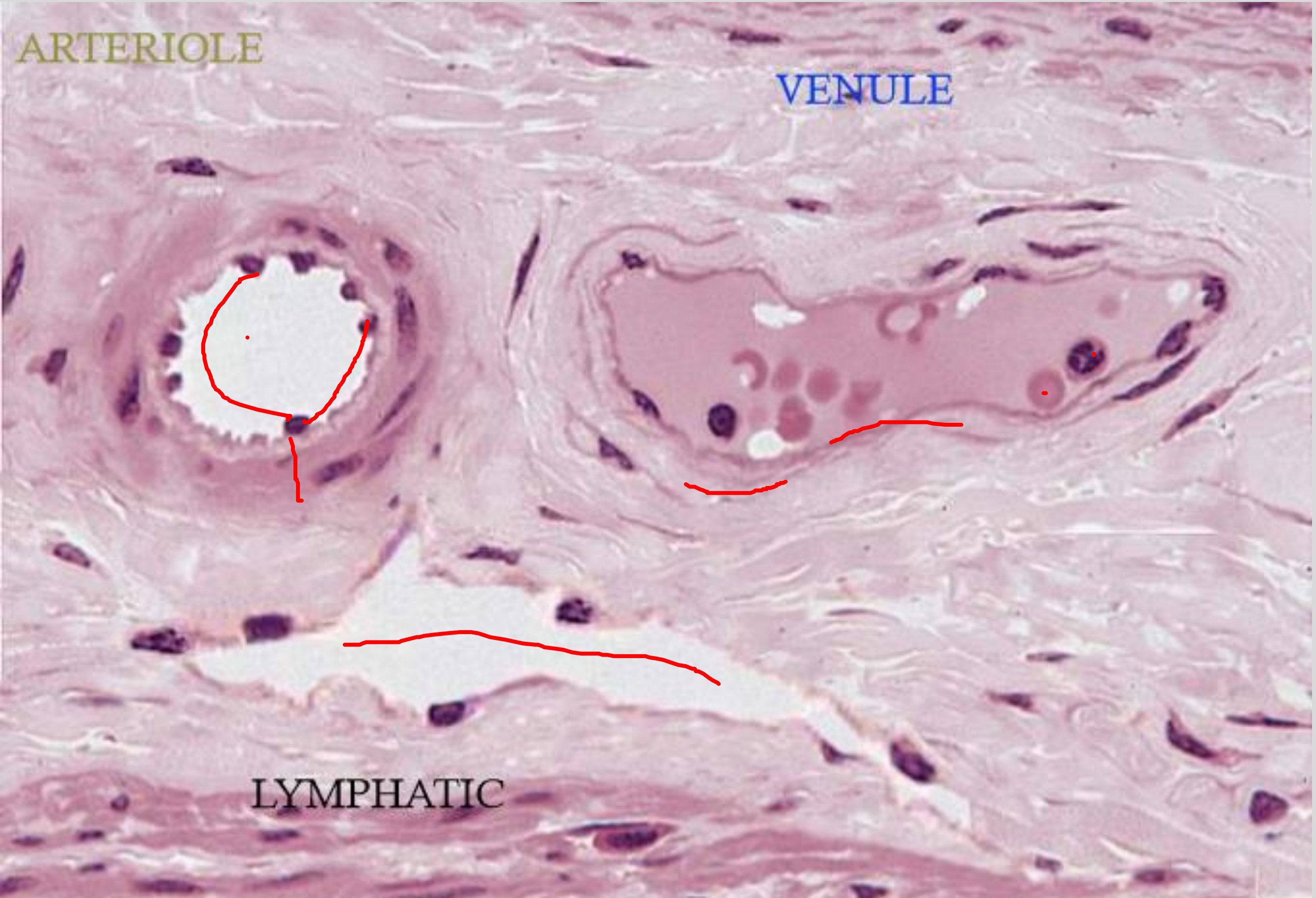
Boundary not clear

t. media: **Elastin**

- **circular**
- **smooth muscle**

t. intima: **endothelium**

Example of arteriole, venule, small lymphatic vessel



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Source: Memorang

VEIN VALVES

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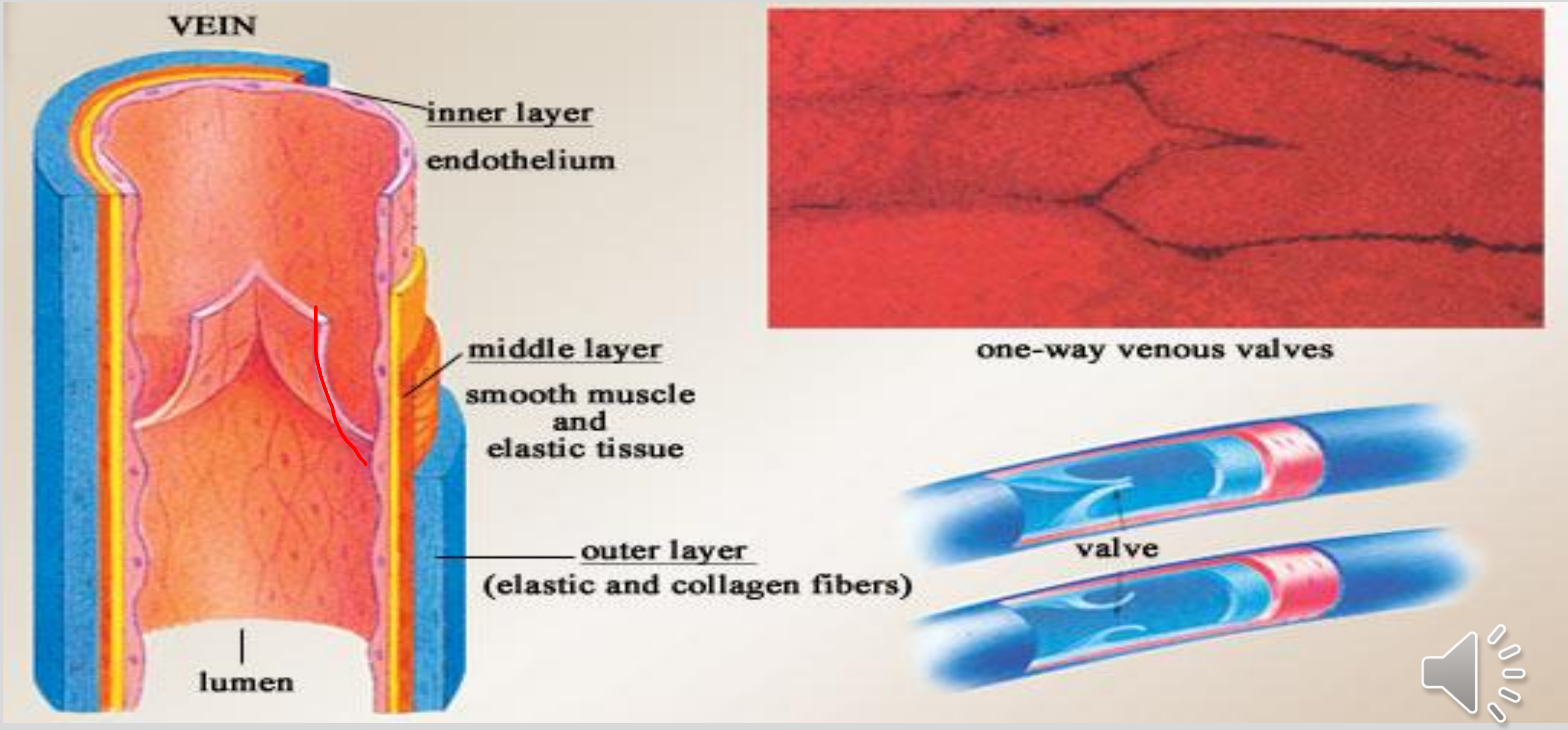

Clinical alert about impaired microvascular blood flow:

Ischemic heart disease: obstructed coronary arteries
Non-Ischemic heart disease: Patent coronary arteries

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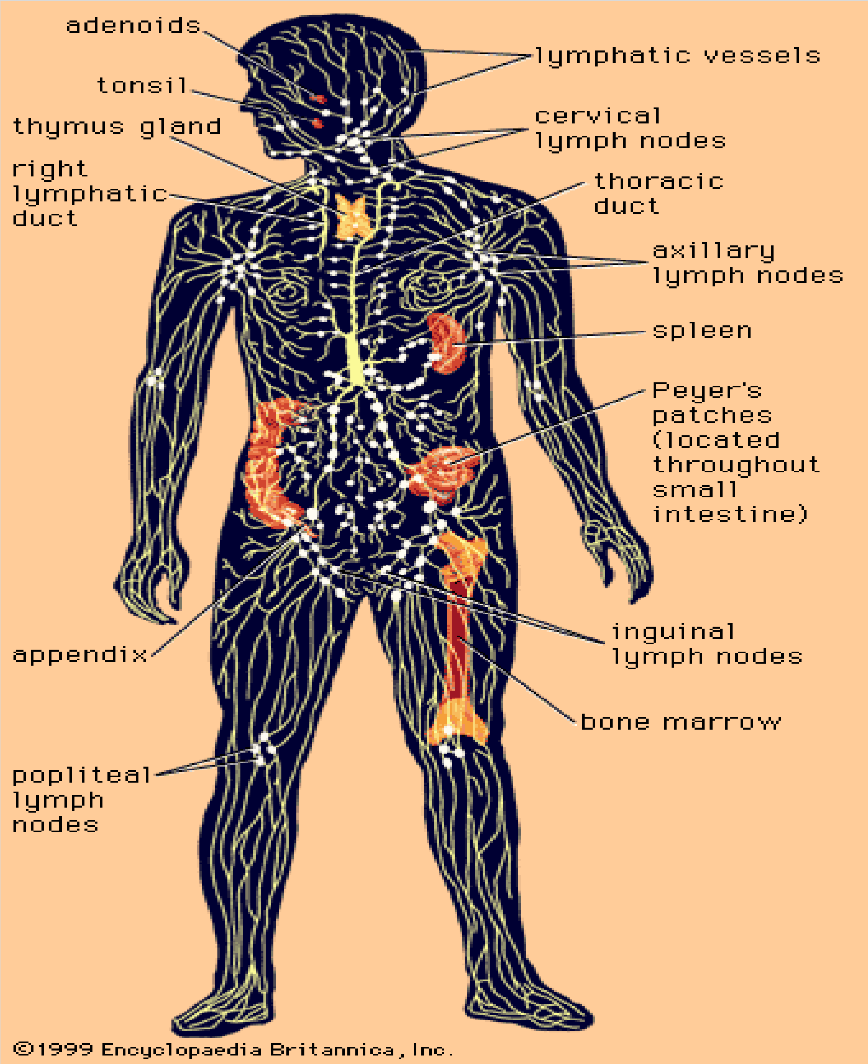
Many recent clinical reports show a high incidence of microvascular blood flow impairment in "Non-Ischemic" heart patients. This includes dilated heart failure (HFrEF), non-dilated heart failure (HFpEF) and diabetic cardiomyopathy. **So, many "non-ischemic" patients may actually have ischemia at the microvascular level**. See pdf review if interested in cardiology.

This has been under reported because the traditional method to measure this (PET scans) uses radioisotopes. So these measurements were rarely collected. Newer, non-radioactive technology is rapidly moving toward increased clinical use.

Animal research from my lab (AM Gerdes) suggests this may be due to reduced levels of cardiac tissue T3 (thyroid hormone), which regulates microvascular blood flow. See pdf.

This will be an important area of clinical research to follow.



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Lymphatics

Lymphatic Vessels

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- Unidirectional, **convey fluid only FROM tissues**
- Smallest are **lymphatic capillaries**, numerous in loose connective tissue under skin epithelium and mucous membranes; **begin as "blind-ended" tubes** in microcapillary beds
- Lymphatic capillaries converge into larger lymphatic vessels which eventually unite to form two main channels, then empty into vascular system by draining into the large veins in the base of the neck
- **Lymph enters the blood at the junctions of the internal jugular and subclavian veins**. The largest lymphatic vessel, the **thoracic duct,** drains most of the body and empties into the veins on the left side. The other main channel is the **right lymphatic trunk.**

Lymphatics, continued

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- **Greater permeability than blood capillaries**, more effective in removing protein-rich fluid from intercellular spaces
- Once fluid enters lymphatic vessel it is known as "**lymph**"
- Also serve to **transport proteins and lipids that are too large** to cross the fenestrations of the absorptive capillaries in the small intestine
- Before **lymph** returns to blood, it **passes through lymph nodes**, where it is exposed to cells of the immune system
- No central pump in lymphatic system. Valves help unidirectional flow and compression of skeletal muscles help move lymph along.

Lymphatics, continued

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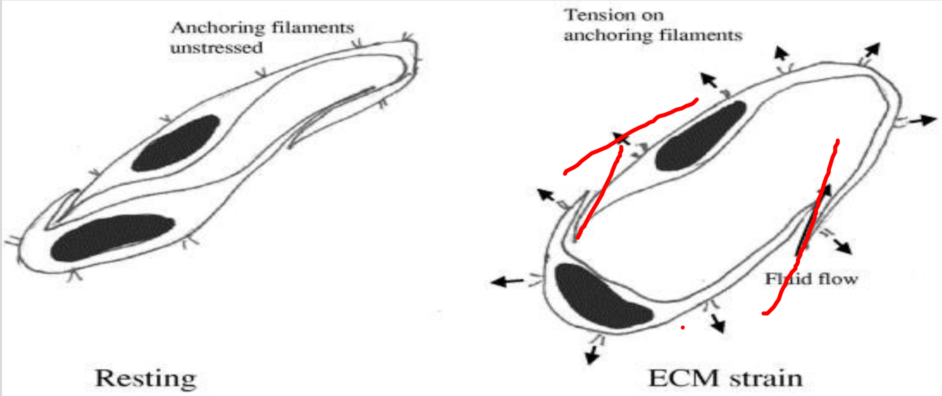

- **Lymphatic capillaries** = tubes of **endothelium which do not have continuous basal lamina** (which is why they are highly permeable) and more variable lumen diameters; **anchoring filaments** extend between the incomplete basal lamina and perivascular collagen to help keep the vessels patent when there is increased tissue pressure (e.g. inflammation)
- Lymphatic vessels progressively become thicker as walls develop more connective tissue and smooth muscle

Lymph vessels

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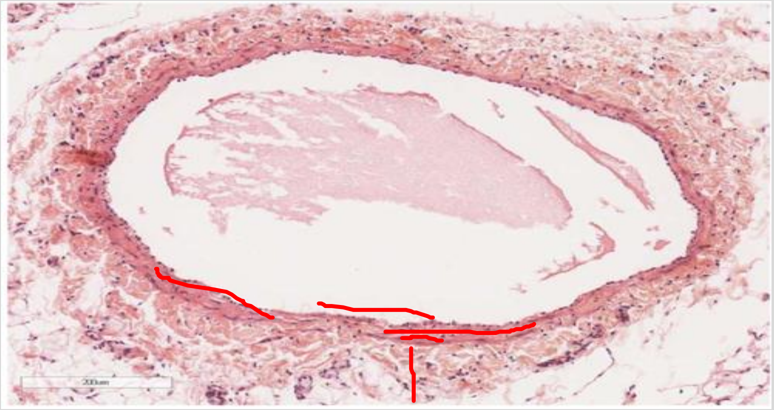

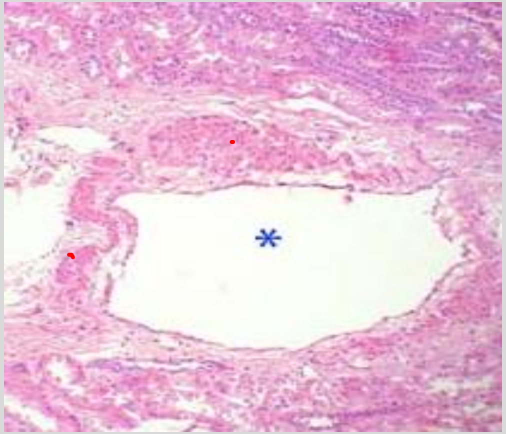

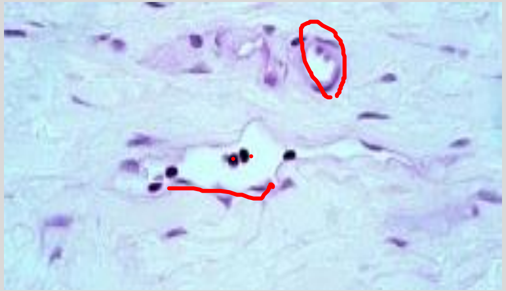

ducts

-Thoracic and right lymphatic ducts empty into neck veins. Larger lymphatic vessels have smooth muscle layer.

larger vessels

-Like medium veins with less developed tunics

-bring lymph through lymph organs

capillaries

-Endothelium with thin connective tissue layer

-dead-end near blood capillaries

-pick up extra-cellular fluid from tissues

Lymph vessels: can have muscle layers and valves

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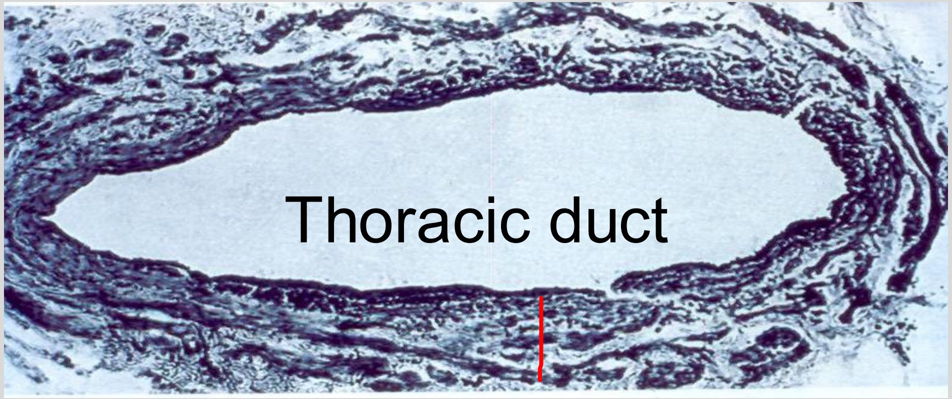
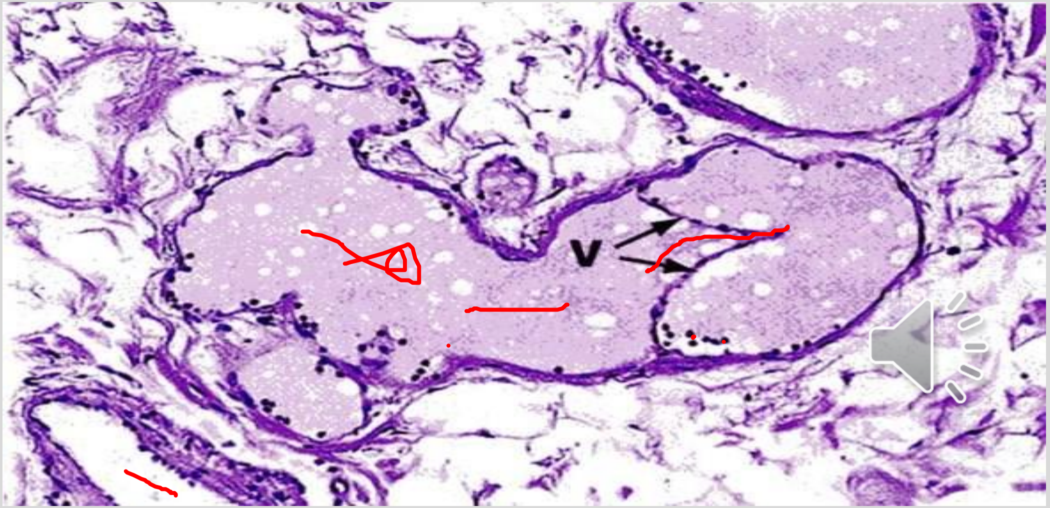


Thoracic duct: note relatively disorganized layers for such a large vessel

Lymphatic vessel showing valve. Note lymphatic fluid coagulated from fixation, presence of WBCs but no RBCs




lymph
vessel
muscular
artery
vein

veinule

lymph vessel
no blood, some endothelial cells

veins

arteriole
smooth muscle in t. media

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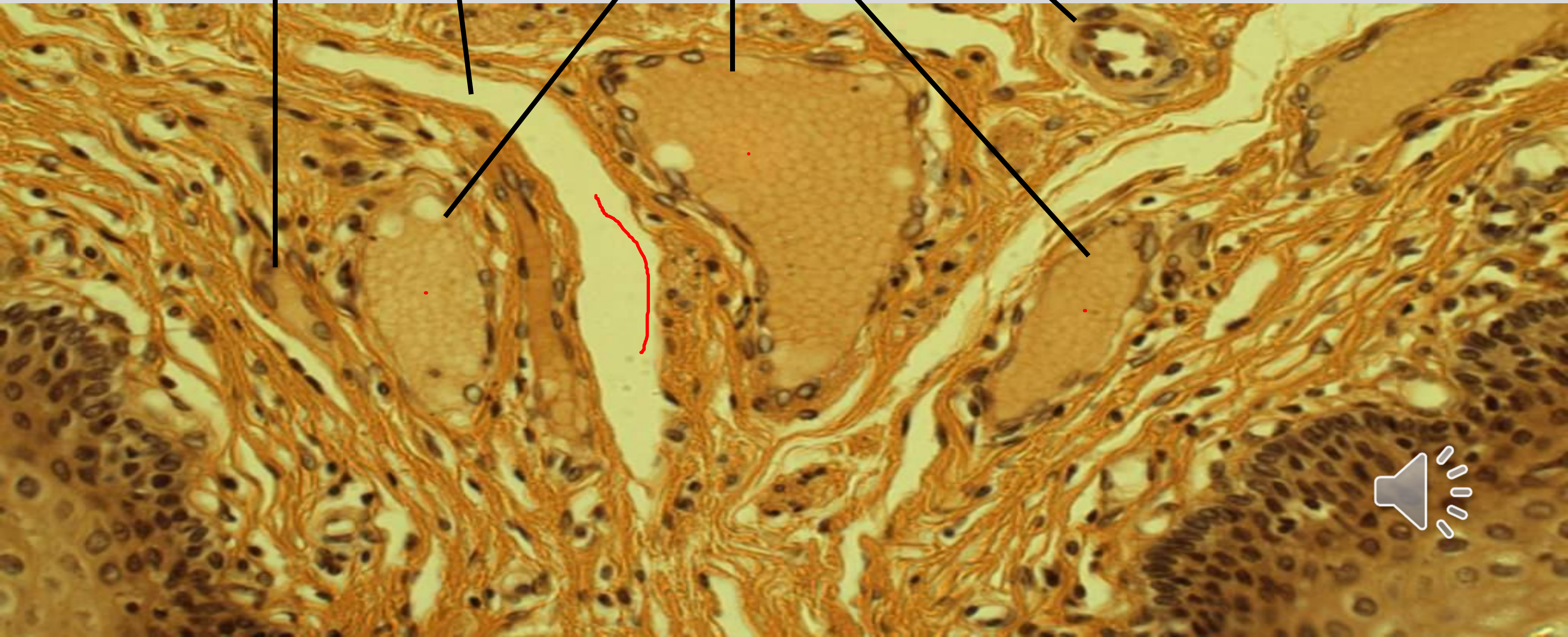

Summary Slide

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- Students will be able to compare and contrast large, medium, and small arteries and veins.
- Students will be able to describe different capillary types and how their morphology varies according to their function.
- Students will be able to discuss the histology and function of lymphatic vessels.
- Students will be able to discuss the unique functions of endothelial cells.
- Students will be able to discuss microvascular components that regulate blood flow.

Source: Histology: A text and Atlas, Ross and Pawlina, Chapter 13, Cardiovascular System, pp 408-429, 434-438

Slide review

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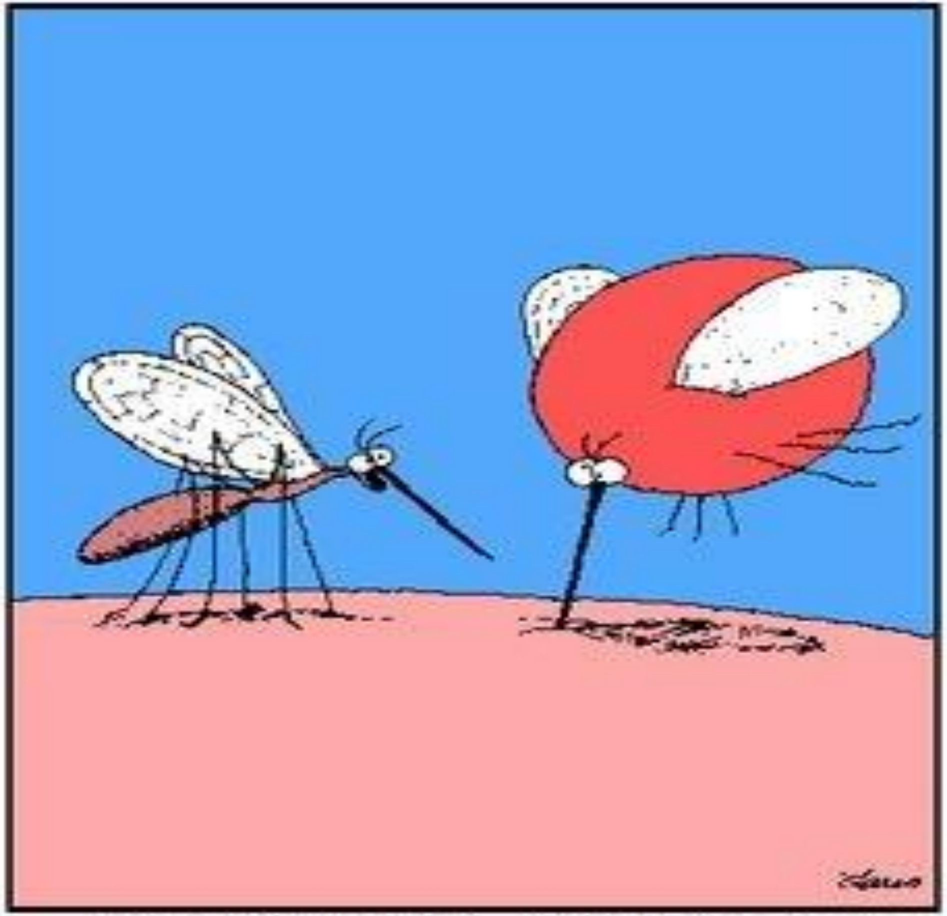
To practice identifying vessels, use this slide from the Univ Michigan site (zoom in and out)

- https://histologyslides.med.umich.edu/Histology/Cardiovascular%20System/098-N_HISTO_40X.htm



"Pull out, Betty! Pull out! . . . You've hit an artery!"

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R